# How to Design a Survey Questionnaire

## & Analyse Results

A Thesis Guide

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2026

## Abstract

The questionnaire survey is one of the most widely used research methods in education, business, health and the social sciences. It allows a researcher to collect comparable information from many people, quickly and at low cost, and to describe patterns across a population. But a survey is only as good as its questionnaire: a badly worded question, a confusing layout or a careless sample can produce results that mislead rather than inform. This guide explains, step by step, how to design a questionnaire, test it, administer it, and analyse the results honestly.

The guide follows the survey process from start to finish. It begins by explaining when a survey is the right method and when it is not, and shows how to turn research questions into questionnaire items. It describes the main types of question — open, closed and Likert-scale items — and gives clear rules for writing good questions: clear, neutral, and about one issue at a time.

The final chapters deal with presenting results in tables and charts, the ethics of survey research, and a complete fictional worked example that shows the whole process in one place, from the first research question to the final table of results. Throughout, the emphasis is practical: each chapter includes worked examples, sample tables and checklists that a student can use while designing and carrying out a real survey. The guide is written for undergraduate and postgraduate students who are running a survey for the first time, and it assumes no previous experience of quantitative research.



Keywords: survey; questionnaire; Likert scale; sampling; response rate; descriptive statistics

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How to use this guide: read Chapters 1 and 2 before you write a single question. Chapters 3 to 7 guide you while you draft and test the questionnaire. Chapters 8 to 10 cover sampling and administration. Chapters 11 to 13 guide the analysis and presentation of results. Chapter 15 is a complete worked example to read alongside your own survey.

## Chapter 1: When a Survey Is the Right Method

A survey is a way of collecting the same information from many people, in a standard form, so that the answers can be counted, compared and summarised. The usual instrument is a questionnaire: a set of written questions that every respondent answers in the same way. Surveys are used to describe attitudes, opinions, knowledge, behaviour and experiences across a population — for example, how satisfied students are with their library, how often teachers use digital tools, or what parents think of a new school timetable. Because the data are standardised, a survey can show patterns that a handful of interviews cannot: how common a view is, who holds it, and how it varies between groups.

A survey is the right method when your research question asks how many, how often, or how much; when you need information from a large number of people; and when the information you need can be captured in short, standard answers. It is also the

right method when you want to compare groups — for example, first-year and final-year students, or teachers in urban and rural schools — because the same questions asked of everyone make the comparison fair. For a student researcher, a survey is often the most practical way to gather a respectable amount of data within a single term.

### 1.1 When a survey is not the right method

A survey is the wrong method when your question asks how or why something happens in depth. A questionnaire can tell you that half of the teachers in a school are unhappy with a new policy; it cannot tell you why they are unhappy, in their own words, with the detail and the history that explain it. For that question you need interviews or a case study. A survey is also wrong when the topic is sensitive or complex in ways that fixed answers cannot capture, or when the people you need to reach cannot read a questionnaire or cannot be contacted.

### 1.2 What a survey can and cannot prove

A well-designed survey describes a population accurately, within limits. It can show that most students prefer morning classes, or that satisfaction is higher among final-year students than among first-years. What it cannot do, on its own, is prove that one thing causes another. If students who attend extra classes also get higher marks, the survey shows a relationship, not a cause: the extra classes may help, or the motivated students may be the ones who attend. A careful researcher reports relationships as relationships, and does not claim causes that the survey cannot support. A survey is a powerful descriptive tool; it is not an experiment, and the report must not pretend otherwise.

[ ] My research question asks how many, how often, or how much.

[ ] The information I need can be captured in short, standard answers.

[ ] I need data from a large number of people, not deep detail from a few.

[ ] I am not asking the survey to prove a cause.

## Chapter 2: From Research Questions to Questionnaire Items

The most common mistake in survey research is to start writing questions before thinking about what the survey must find out. The result is a long questionnaire full of interesting questions that do not answer the research question, and missing the questions that would. The correct order is the reverse: start with the research question, break it into the pieces of information you need, and only then write a question for each piece. Every item in the questionnaire should earn its place by serving the research question. If an item does not, delete it, however interesting it is.

### 2.1 Breaking the question into variables

Begin by writing the research question as one sentence. Then list the variables — the things you need to measure — that the question contains. A question about student

satisfaction with the library, for example, contains several variables: how often students use the library, what they use it for, how satisfied they are with the books, the opening hours, the staff, and the study space, and background facts such as their year of study and subject. Write each variable on a separate line. This list is the skeleton of the questionnaire, and it stops you from forgetting a key variable or including an irrelevant one. Show the list to your supervisor before you write any items: it is far cheaper to fix the list than to rewrite a finished questionnaire.

## 2.2 One item per variable, or several?

Some variables can be measured with a single question: 'How often do you visit the library?' is one item. Others need several questions, because they are made of parts. Satisfaction with the library is such a variable: a single question — 'Are you satisfied with the library?' — is too blunt, because a student may love the books and hate the opening hours. Measuring satisfaction properly means asking separately about the books, the hours, the staff, the space and the computers, and then combining the answers.

## 2.3 Background questions

Every survey needs a small set of background questions — sometimes called demographic items — so that the results can be broken down by group: year of study, subject, gender, age group, or whatever matters for your research question. Keep these few and relevant. Ask only for background information that you will actually use in the analysis, and that respondents will be willing to give. Do not ask for a name, and do not ask for details that could identify a person in a small group. Place background questions at the end of the questionnaire, where they feel natural, unless your analysis depends on grouping respondents from the start.

**Worked Example: From question to items**

Research question: 'How satisfied are students with their university library?' Variables: frequency of use, purpose of use, satisfaction with books, hours, staff, space and computers, and background (year, subject). Items: 'How often do you visit the library?' (frequency); 'What do you mainly use the library for?' (purpose); five satisfaction items, one for each aspect; and two background items at the end. Every item traces back to the research question, and no item is there by accident.

- [ ] My research question is written as one sentence.
- [ ] I have a written list of the variables the question contains.
- [ ] Each variable has at least one questionnaire item.
- [ ] Background questions are few, relevant and placed at the end.

# Chapter 3: Types of Questions: Open, Closed, Likert Scales

Questionnaire items come in three main types, and a good questionnaire usually uses all three. Closed questions offer fixed answers to choose from; open questions let the respondent answer in their own words; Likert-scale items ask the respondent to place themselves on a scale, usually from agreement to disagreement. Each type has strengths and weaknesses, and the choice of type should follow from what the item needs to measure. This chapter explains each type, with examples, so that you can choose deliberately rather than by habit.

## 3.1 Closed questions

A closed question gives the respondent a list of answers and asks them to choose one, or more than one. 'How often do you visit the library?' with the options Daily, Several times a week, Weekly, Monthly, Rarely, Never is a closed question. Closed questions are easy to answer, quick to complete, and easy to analyse: the answers can be counted directly. Their weakness is that they limit the respondent to the answers you thought of. If the true answer is not on the list, the respondent must choose the nearest one, or give up. Guard against this by making the list complete — add an Other option with space to write in the answer — and by making the options cover all possibilities without overlapping. 'Aged 20-30' and 'Aged 30-40' overlap at 30; 'Aged 20-29' and 'Aged 30-39' do not.

## 3.2 Open questions

An open question has no fixed answers: 'What one change would most improve the library?' The respondent writes whatever they wish. Open questions are valuable because they can reveal answers the researcher never thought of, and they give the respondent a voice. Their weakness is that they take longer to answer, some respondents skip them, and the answers must be read and coded by hand before they can be summarised. Use open questions sparingly — one or two in a questionnaire, usually at the end — and use them where the range of possible answers is genuinely unknown. Do not use an open question where a closed question would do, and do not use one for information you will need to count precisely.

## 3.3 Likert-scale items

A Likert item presents a statement and asks the respondent to say how far they agree with it, on a scale. The classic form has five points: Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree. Likert items are the workhorse of attitude surveys, because they measure strength of feeling, not just direction, and because a set of items on the same topic can be combined into an overall score. Write each statement as a single clear claim, in plain language, and keep the scale the same throughout the questionnaire so that respondents do not have to relearn it.

| Type | Best for | Main weakness |
|---|---|---|

| | | |
|---|---|---|
| Closed question | Facts, frequencies, choices | Limits answers to the list given |

| | | |
|---|---|---|
| Open question | New ideas, reasons in own words | Slow to answer; needs hand coding |

| | | |
|---|---|---|
| Likert-scale item | Attitudes and strength of feeling | Needs careful wording; easy to tick without thinking |

[ ] Most items are closed or Likert-scale; open items are few.

[ ] Closed options cover all possibilities and do not overlap.

[ ] Each Likert statement is a single clear claim.

[ ] The response scale is the same throughout the questionnaire.

# Chapter 4: Writing Good Questions

The quality of a survey depends on the quality of its questions. A question that is unclear, leading, or about two things at once will produce answers that cannot be trusted, and no analysis can fix them. The rules in this chapter are simple, but they must be applied to every item, without exception. Write each question, then read it as a respondent would: could it be misunderstood? Could it be answered in two different ways? Does it push the respondent towards one answer? If the answer to any of these is yes, rewrite it.

## 4.1 Clear and simple language

Use short sentences and everyday words. Avoid jargon, abbreviations and technical terms unless you are certain that every respondent understands them. 'How satisfied are you with the provision of learning resources?' is unclear to many students; 'How satisfied are you with the books and computers in the library?' is clear. Avoid vague words such as 'often', 'regularly' and 'sometimes', because different people mean different things by them: give a concrete scale instead, such as 'every day', 'several times a week', 'once a week', 'less than once a week'.

## 4.2 One issue at a time

A double-barrelled question asks about two things at once: 'How satisfied are you with the library books and the opening hours?' A respondent who loves the books and hates the hours cannot answer it honestly, and the answer you receive is meaningless. Split every double question into two separate items. The same rule applies to questions that hide an assumption: 'Why do you rarely visit the library?' assumes that the respondent rarely visits it. Ask first whether they visit, and only then ask why. Read each question and count the issues it raises; if the count is more

than one, split it.

### 4.3 Neutral wording

A leading question pushes the respondent towards an answer: 'Do you agree that our excellent new library is a great improvement?' The words 'excellent' and 'great' tell the respondent what the researcher wants to hear. Write the question neutrally: 'How satisfied are you with the library?' The same applies to questions that carry a judgement: 'Do you waste time on social media?' will produce defensive answers, while 'How many hours a day do you spend on social media?' is neutral and will produce better data.

**Worked Example: Weak and strong questions**

Weak: 'Do you agree that the new timetable, which most students love, is better than the old one?' It is leading, it assumes the answer, and it asks about two timetables at once. Strong: 'How satisfied are you with the current timetable?' followed by a five-point scale from Very satisfied to Very dissatisfied. The strong version is neutral, asks one thing, and gives the respondent a clear scale on which to answer.

- [ ] Every question uses short sentences and everyday words.
- [ ] No question asks about two things at once.
- [ ] No question contains a word that praises or blames.
- [ ] Vague words like 'often' are replaced by concrete options.
- [ ] Each question can be answered honestly by every respondent.

## Chapter 5: Question Order and Layout

The order of the questions and the layout of the page matter more than most beginners expect. A questionnaire that starts with a difficult question, jumps between topics, or looks crowded will lose respondents, and the answers it does receive will be worse. The rules are simple: start easy, group related questions together, keep a logical flow, and make the page look clean and inviting. Think of the questionnaire from the respondent's point of view: they have agreed to give you ten minutes, and every unnecessary difficulty spends their goodwill.

### 5.1 Order: from easy to sensitive

Begin with questions that are easy, interesting and non-threatening: how often the respondent uses a service, or what they think of a familiar topic. These early questions build confidence and momentum. Place the most important questions in the first half of the questionnaire, while the respondent is still fresh and attentive. Place background questions — age, year of study, income — at the end, where they feel natural and where a respondent who dislikes them can skip them without abandoning the whole survey.

### 5.2 Grouping and flow

Group questions by topic, and give each group a short heading: 'About the library',

'About your studies', 'About you'. A heading tells the respondent that the topic has changed and helps them switch their thinking. Within a group, keep the order logical: general questions before specific ones, and questions about behaviour before questions about opinion. Avoid jumping back and forth between topics, which confuses respondents and makes the questionnaire feel disorganised. If a question only applies to some respondents — for example, only to those who use the library — use a filter instruction: 'If you never visit the library, please go to Question 12.' Test every filter in the pilot, because a wrong filter loses data.

### 5.3 Layout on paper and on screen

On paper, use a clear font, generous spacing, and plenty of white space. Number every question, keep the response options aligned, and do not split a question across two pages. A crowded questionnaire looks like hard work, and respondents will treat it as hard work. On screen, the same principles apply: one question per screen or a short group per screen, large touch targets, and a progress indicator so that respondents can see how far they have come. Test the layout on a phone as well as on a computer, because many respondents will answer on a phone.

[ ] The first questions are easy and interesting.

[ ] The most important questions are in the first half.

[ ] Questions are grouped by topic with short headings.

[ ] Filters are written clearly and tested in the pilot.

[ ] The layout is clean, with white space and aligned options.

## Chapter 6: Scales and Response Options

A response scale is the set of answers a respondent can choose from, and its design shapes the data. A scale with too few points cannot capture differences; a scale with too many points confuses respondents; a scale without a middle point forces people who are undecided to take a side they do not hold. This chapter explains how to choose scales and response options that are balanced, complete and easy to use.

### 6.1 Choosing the number of points

For attitude questions, a five-point scale is the standard choice and a safe default for a student survey: Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree. Five points give enough room to show strength of feeling without overwhelming the respondent. A four-point scale, with no middle point, forces a choice and is sometimes used when the researcher wants to avoid a pile of neutral answers — but it can also force false answers from people who genuinely have no view, so use it only with a clear reason. Whichever scale you choose, use it consistently throughout the questionnaire.

### 6.2 Balanced and labelled options

A scale must be balanced: the same number of positive and negative options, with a neutral middle. 'Excellent, Good, Fair, Poor' is unbalanced, because it has two positive

options and one negative one; 'Excellent, Good, Fair, Poor, Very poor' is balanced. Label every point on the scale, not just the ends: respondents interpret numbers differently, and a scale labelled only at the ends — 1 and 5 — will be used inconsistently. For frequency questions, label the options with concrete words: 'Every day, Several times a week, Once a week, Less than once a week, Never' is clear and covers the full range. Check that the options are in a sensible order, from most to least, and that they do not overlap.

### 6.3 Avoiding common scale errors

Four errors appear again and again in student questionnaires. First, the unbalanced scale, described above. Second, the missing option: a question about employment that offers 'Full-time' and 'Part-time' but not 'Not employed' leaves some respondents with no honest answer. Always include the full range, and add 'Other' or 'Not applicable' where needed. Third, the overlapping option, such as age bands that share a boundary. Fourth, the double scale: asking 'How satisfied are you?' and offering the answers 'Very satisfied, Satisfied, Dissatisfied, Very dissatisfied, Never use the library' mixes satisfaction with use, and the last option does not belong on that scale. Keep one scale per question, and use a filter for respondents to whom the question does not apply.

**Worked Example: A balanced scale**

Poor: 'How good is the library? Excellent / Good / Poor.' It has two positive options, one negative option, and no middle. Good: 'How would you rate the library? Excellent / Good / Fair / Poor / Very poor.' It is balanced, every point is labelled, and the middle point gives undecided respondents an honest answer.

- [ ] My attitude scale has five points, with a neutral middle.
- [ ] The scale is balanced: equal positive and negative options.
- [ ] Every point on the scale is labelled in words.
- [ ] Options cover the full range and do not overlap.
- [ ] The same scale is used consistently through the questionnaire.

## Chapter 7: Piloting the Questionnaire

Piloting means testing the questionnaire on a small group of people like your real respondents, before the main survey begins. It is the single most valuable step in survey research, and skipping it is the most expensive mistake a student can make: a flaw found in the pilot costs an afternoon to fix, while the same flaw found after the main survey has run can ruin the whole study. Pilot with ten to twenty people if you can, and treat the pilot as a serious test, not a formality.

### 7.1 What the pilot is for

The pilot has four jobs. First, it finds questions that are misunderstood: if several pilot respondents ask what a question means, the question is unclear. Second, it finds questions that cannot be answered: options that are missing, scales that do not fit, or

filters that send people to the wrong place. Third, it measures the time the questionnaire takes, so that you can give respondents an honest estimate. Fourth, it tests the whole process: whether the online form works on a phone, whether the paper copies print clearly, and whether the data come back in a form you can analyse.

### 7.2 How to run a pilot

Choose pilot respondents who resemble your real respondents, and ask them to complete the questionnaire exactly as a real respondent would, without your help. Watch the time it takes, and note where they hesitate, frown, or ask a question. Afterwards, interview them briefly: Which questions were unclear? Which were hard to answer? Was anything missing? Was the questionnaire too long? Did anything feel intrusive? Write down everything they say, and revise the questionnaire in response. Then pilot the revised version on a few more people, to check that the fixes worked and that nothing new has broken. Two rounds of piloting are better than one, and the time they take is repaid many times over.

### 7.3 Revising after the pilot

Revise boldly after the pilot. Delete questions that do not work, rewrite questions that confuse people, add options that are missing, and shorten the questionnaire if pilot respondents found it too long. Keep a note of what you changed and why, so that you can describe the piloting in your report. Do not include pilot data in the main results: the pilot questionnaire was different from the final one, and the pilot respondents are not part of the main sample. A short paragraph in the methodology section, describing the pilot and the changes it led to, shows the examiner that the questionnaire was tested rather than assumed to be right.

[ ] The questionnaire was piloted with people like the real respondents.

[ ] I timed the pilot and recorded where respondents hesitated.

[ ] I interviewed pilot respondents about unclear or missing questions.

[ ] The questionnaire was revised after the pilot and tested again.

[ ] Pilot data are not included in the main results.

## Chapter 8: Sampling and Sample Size

A survey rarely reaches everyone in the population it describes. Instead, the researcher studies a sample — a smaller group chosen to represent the population — and uses the sample's answers to describe the population. The quality of the survey depends on how the sample was chosen and how large it is. A well-chosen sample can describe a population accurately; a badly chosen one can mislead, however many people answer. This chapter gives qualitative guidance on sampling and sample size. It does not invent formulas or percentage claims: for the exact number you need, consult a standard sample-size table or calculator for your population, and discuss it with your supervisor.

## 8.1 Defining the population

Begin by defining the population precisely: the whole group your study is about. 'Students at my university' is a start, but it is usually too broad for a student project. Narrow it to something you can actually reach: 'undergraduate students in the Faculty of Education at my university, in the current academic year'. Write the definition down, because it decides who is in the sample and who is not. If your population is small — a single department, for example — you may be able to survey everyone, which is called a census, and sampling is not needed. For larger populations, you will need a sample.

## 8.2 Random and non-random sampling

In random sampling, every member of the population has a known chance of being selected, and the selection is made by chance rather than by choice. The common forms are simple random sampling, where names are drawn at random from a list, and stratified sampling, where the population is divided into groups — by year of study, for example — and a random sample is drawn from each group. Random sampling is the best way to get a representative sample, because it removes the researcher's choice from the selection. Non-random sampling includes convenience sampling, where the researcher surveys whoever is available, and purposive sampling, where the researcher deliberately chooses people with particular characteristics. Convenience sampling is common in student projects because it is practical, but it must be named honestly as convenience sampling, and its limits must be stated: the people who happened to be available may differ from the population as a whole, and the results cannot be generalised with the same confidence.

## 8.3 How large should the sample be?

Larger samples generally give more accurate results than smaller ones, because sampling error — the difference between the sample's answers and the population's true answers — tends to shrink as the sample grows. The right size depends on the size of your population, how precise your results need to be, and what is practical in the time you have. Do not invent a number or copy one from another study: use a standard sample-size table or an established sample-size calculator, enter your population size and the precision you need, and follow its guidance. Discuss the result with your supervisor, and report the target sample size, the method used to choose it, and the number of responses actually received in your report.

## 8.4 Reaching the sample

A good sampling plan fails if the people selected cannot be reached. Plan the practical side carefully: how will you contact them, how many reminders will you send, and what will you do if someone cannot be reached? Keep a record of how many people were invited, how many responded, and how many could not be contacted. This record is the basis of the response rate, which is discussed in Chapter 10, and it is essential evidence for the reader who wants to judge how representative your sample is.

| Method | How it works | Main strength and limit |
|---|---|---|
| Simple random | Names drawn by chance from a list | Fair; needs a complete list |
| Stratified random | Random sample from each subgroup | Guarantees each group is included |
| Convenience | Whoever is available | Practical; may not represent the population |
| Purposive | Chosen for particular characteristics | Good for special groups; not representative |

[ ] My population is defined precisely, in writing.

[ ] My sampling method is named and justified.

[ ] The target sample size comes from a standard table or calculator, not from guesswork.

[ ] I have a plan for contacting the people selected.

## Chapter 9: Administering the Survey

Administering a survey means getting the questionnaire to the respondents and getting the answers back, in a form you can analyse. The three main ways are paper, online and face-to-face, and each has its strengths and weaknesses. The choice depends on who your respondents are, how many there are, and what resources you have. Whichever way you choose, plan it carefully, keep a record of what you did, and treat the respondents' time with respect.

### 9.1 Paper surveys

Paper questionnaires are useful where respondents do not have reliable internet access, or where the survey is given in a classroom or a waiting room. They are cheap to produce and easy to hand out, but the answers must be entered into a computer by hand afterwards, which takes time and can introduce errors. Number every copy, keep a log of how many were given out and how many came back, and enter the data as soon as possible, checking a sample of entries against the originals.

### 9.2 Online surveys

Online surveys are the most common choice for student projects, because they are fast, cheap, and the data arrive already in a spreadsheet. Choose a survey tool that your university provides or that you know well, and test the questionnaire on a phone

and on a computer before sending it out. Send the invitation with a short, honest message: what the survey is about, how long it takes, that participation is voluntary, and that answers are anonymous. Send one or two reminders, spaced a few days apart, and then stop: repeated reminders annoy people and rarely add useful responses.

### 9.3 Face-to-face surveys

In a face-to-face survey, the researcher reads the questions to the respondent and records the answers. This method reaches people who cannot read a questionnaire, and it usually produces a high response rate, because it is hard to refuse a polite person standing in front of you. Its weakness is cost: it takes far longer than the other methods, and the researcher's presence can influence the answers — respondents may give the answer they think the researcher wants. If you use this method, read the questions exactly as written, do not explain or rephrase them on the spot, and record the answer the respondent gives. Train any helpers to do the same, and check their work early.

[ ] The method fits my respondents: paper, online or face-to-face.

[ ] The questionnaire has been tested on the device respondents will use.

[ ] The invitation states the purpose, the time needed, and that participation is voluntary.

[ ] I keep a log of questionnaires sent, returned and entered.

## Chapter 10: Response Rates and Non-Response

The response rate is the proportion of the people you invited who actually answered. It matters because the people who answer may differ from the people who do not. If the satisfied students answer and the dissatisfied ones throw the questionnaire away, your results will look better than the truth. A high response rate reduces this risk; a low one increases it. You cannot eliminate non-response, but you can reduce it, measure it, and report it honestly.

### 10.1 Raising the response rate

Response rates vary a great deal by method and by population, and no single figure can be promised in advance. Several practical steps help. Keep the questionnaire short: a survey that takes five minutes will be answered far more often than one that takes twenty. Make the invitation personal and clear: explain why the survey matters and who will use the results. Send reminders, but no more than two. Choose a good time: students answer more readily during term time than during examinations. And make the questionnaire easy: a clean layout, clear instructions, and no question that makes the respondent stop and think about whether to continue. Small courtesies — a thank-you at the end, a summary of the results offered to respondents — also help, and cost nothing.

### 10.3 Reporting the response rate

Report the response rate plainly, with the numbers: how many people were invited, how many answered, and how many could not be contacted. Do not hide a low response rate, and do not describe it with vague words such as 'good' or 'satisfactory'. State the rate, describe what you did to raise it, and discuss honestly what non-response might mean for your findings. A report that says '120 of the 200 students invited responded, a response rate of 60 per cent' is honest and useful; a report that gives no numbers at all invites the examiner to assume the worst.

**Worked Example: Reporting a response rate**

Weak: 'Many students completed the survey.' The reader learns nothing. Strong: 'The questionnaire was sent to 200 undergraduate students; 120 completed it, a response rate of 60 per cent. Final-year students were under-represented: they were 30 per cent of the population but only 18 per cent of respondents, so the findings should be read with that limit in mind.' The strong version gives the numbers, names the gap, and states its consequence.

[ ] The questionnaire is short enough to finish in a few minutes.

[ ] I will send at most two reminders.

[ ] I keep a count of invited, answered and unreachable people.

[ ] The report states the response rate with numbers, not vague words.

# Chapter 11: Cleaning the Data

Before any analysis begins, the data must be cleaned: checked for errors, missing answers and impossible values, and prepared for analysis. Data cleaning is unglamorous work, but it is essential. An analysis built on dirty data produces results that look precise and are wrong, and the errors are often invisible until an examiner asks an awkward question. Budget real time for cleaning — often a full day for a survey of a few hundred responses — and keep a record of every decision you make.

## 11.1 Checking for errors

Begin by checking that the data were entered correctly. If the data were entered by hand, check a sample of entries against the original questionnaires. If the survey was online, check that the questions and the answers match: a question about age should not contain the answer 'Strongly agree'. Look for impossible values: an age of 200, a rating of 9 on a five-point scale, or a date in the future. Look for duplicates: the same respondent answering twice, which sometimes happens online. Look for patterns that suggest careless answering, such as a respondent who ticked the same box for every question, or who finished a twenty-minute questionnaire in thirty seconds.

## 11.2 Missing data

Some respondents will skip some questions. Small amounts of missing data are normal and are usually handled by analysing each question using the answers it received, and reporting the number of answers for each question. Larger amounts need thought. If a whole section is missing for many respondents, there may be a

fault in the questionnaire — a filter that sent people to the wrong place, for example — and you should investigate before analysing. Never fill in missing answers with guesses, and never delete a respondent's whole record because of one missing answer unless you have a clear reason and you say so. Report the amount of missing data for the main questions, so that the reader can judge its effect.

### 11.3 Coding open answers

Open questions produce text, and text must be coded before it can be counted. Read all the answers to an open question, group them into a small number of categories — usually five to eight — and assign each answer to a category. Give each category a clear name and a definition, and keep a note of the answers that did not fit any category. Count how many answers fall in each category, and quote one or two typical answers to illustrate each one. Coding is a judgement, so ask a second person to code a sample of the answers and compare the results; if you disagree often, the categories need to be clearer. Report the categories and the counts, and keep the original answers so that the reader can check them.

[ ] Hand-entered data have been checked against the originals.

[ ] Impossible values and duplicates have been found and dealt with.

[ ] Missing data are reported for the main questions.

[ ] Open answers are coded into clear categories, with a second coder checking a sample.

[ ] Every cleaning decision is recorded.

## Chapter 12: Analysing Results: Descriptive Statistics and Cross-Tabulation

Analysis is the stage where the cleaned data become findings. For most student surveys, the analysis is descriptive: it summarises what the respondents said, using counts, percentages, averages and simple comparisons between groups. This chapter explains the main descriptive techniques and how to use them honestly. It does not require advanced statistics, and it does not invent formulas: the techniques here are the standard ones taught in introductory statistics courses, and any statistics textbook or your supervisor can guide you on the details.

### 12.1 Counts, percentages and averages

Begin with the basics. For each closed question, count how many respondents chose each option, and express the counts as percentages of the people who answered that question. Always state the number of answers the percentage is based on: '65 per cent of the 120 respondents' is informative; '65 per cent' on its own is not. For numerical data, such as age or hours of study, report the average and the range — the lowest and highest values — so that the reader can see how spread out the answers are. Keep the presentation simple: the reader should be able to grasp each result in a few seconds.

## 12.2 Cross-tabulation

Cross-tabulation is the most useful technique in survey analysis. It compares the answers of different groups: for example, satisfaction with the library among first-year students compared with final-year students, or library use among students in different subjects. The result is a table with one group down the side and the other across the top, showing the percentage in each cell. Cross-tabulation turns a flat description into a finding: it shows not only that half the students are satisfied, but that satisfaction is higher among final-year students than among first-years. Choose the comparisons that follow from your research question, and do not cross-tabulate everything with everything — a report full of tables that answer no question is worse than a short report that answers the question well.

## 12.3 What not to claim

Descriptive statistics describe your sample; they do not, on their own, prove that a difference between groups is real rather than a chance result of sampling. If you want to test whether a difference is likely to be real, use a standard statistical test appropriate to your data — your supervisor or a statistics textbook can advise which one — and report the result plainly. Report the numbers, describe the pattern, and let the reader see how strong the evidence is.

| Technique | What it shows | When to use it |
|---|---|---|
| Frequency count | How many chose each option | For every closed question |
| Percentage | The share of each option | To compare questions with different numbers of answers |
| Average and range | The typical value and the spread | For numerical data such as age or hours |
| Cross-tabulation | How answers differ between groups | To compare groups named in the research question |

- [ ] Every percentage states the number of answers it is based on.
- [ ] Cross-tabulations follow from the research question.
- [ ] No claim of significance is made without a statistical test.
- [ ] Small differences are described as small, not exaggerated.

# Chapter 13: Presenting Results in Tables and Charts

Results are only useful if the reader can understand them quickly, and that depends on how they are presented. A good table or chart lets the reader grasp a finding in seconds; a bad one hides it behind clutter, misleading scales or missing labels. This chapter gives the rules for tables and charts that present survey results clearly and honestly. The same rules apply in a thesis, a report or a presentation.

## 13.1 Tables

Use a table when the reader needs exact numbers, or when several numbers must be compared at once. Give every table a number and a clear title that says what it shows: 'Table 13.1: Satisfaction with library services, by year of study (per cent)'. Label every row and column, state the number of respondents, and give percentages to one decimal place at most — more precision is false precision. Keep the table simple: no more than six or seven columns, no unnecessary lines, and the most important comparison easy to find. In the text, refer to the table by number and point out the one or two findings that matter most. Do not repeat every number in the text; the table already contains them.

## 13.2 Charts

Use a chart when the reader needs to see a pattern: a trend, a comparison, or a distribution. A bar chart is the best choice for comparing groups — satisfaction among first-years, second-years and final-years, for example — because the lengths of the bars can be compared at a glance. A pie chart suits a single question with a few options, such as the share of students who use the library daily, weekly or rarely, but it becomes unreadable with more than five or six slices. Start the scale of a bar chart at zero, so that the bars are honest: a chart that starts at 50 per cent makes a small difference look enormous. Label every axis, give the chart a title, and state the number of respondents.

## 13.3 Writing about results

Write about each result in three sentences: what the table or chart shows, what it means, and what its limit is. 'Table 13.1 shows that satisfaction with opening hours is higher among final-year students (78 per cent) than among first-year students (52 per cent). This suggests that final-year students, who use the library more intensively, value the evening hours that first-years rarely use. The difference is based on 120 respondents, so it should be read as a pattern in this sample rather than a proven difference in the whole population.'

| Year of study | Satisfied with hours | Satisfied with books | Number of respondents |
|---|---|---|---|
| First year | 52% | 68% | 45 |
| Second year | 61% | 70% | 40 |

| | | | |
|---|---|---|---|
| Final year | 78% | 75% | 35 |
| All students | 63% | 71% | 120 |

[ ] Every table and chart is numbered, titled and referred to in the text.

[ ] Bar charts start at zero; pie charts have at most six slices.

[ ] No data appear in both a table and a chart.

[ ] The text points out the main finding and its limit for each result.

# Chapter 14: Ethics

Survey research involves real people, and it carries ethical duties that must be met before a single questionnaire is sent out. Most universities require ethics approval before the survey begins. Apply for it early, describe the survey honestly, and do not begin collecting data until approval is granted. An ethics section is a required part of the report, not an optional extra, and examiners treat a missing one as a serious fault.

## 14.1 Informed consent and voluntary participation

Every respondent must understand what the survey is about, how long it will take, what the data will be used for, and that participation is voluntary: they may skip any question, and they may stop at any time without giving a reason and without any penalty. State this in the introduction to the questionnaire, in plain language, and make it genuinely true — do not require answers to every question in an online form, and do not pressure anyone to take part. Where the survey involves children, consent must come from a parent or guardian as well as from the child, and the school or organisation must also agree.

## 14.2 Anonymity and confidentiality

Do not ask for names, and do not collect details that could identify a person, especially in a small group where a combination of background answers — one final-year student in a particular subject, for example — could give someone away. If the survey is online, check that the tool does not record email addresses or internet addresses unless you have a clear reason and the respondents' consent. Store the data securely, do not share individual responses with anyone outside the study, and report results only in groups large enough that no individual can be recognised.

## 14.3 Honesty in reporting

Report the survey honestly: the response rate, the sampling method, the questions as they were asked, and the results as they were found, including the results that do not support your expectations. Do not drop awkward questions from the report, do not reword questions after the event, and do not present the results of a convenience sample as if they described the whole population. Keep the questionnaire and the raw data so that your supervisor can check them. A survey reported honestly, with its

limits stated, is good research; a survey reported selectively is not.

[ ] Ethics approval was obtained before the survey began.

[ ] The questionnaire states that participation is voluntary.

[ ] No names or identifying details are collected.

[ ] Results are reported only in groups, never for individuals.

[ ] The response rate and sampling method are reported honestly.

# Chapter 15: Worked Example: A Complete Fictional Survey

This chapter presents a complete survey in outline, so that you can see how the parts of this guide fit together. The survey is entirely fictional: the university, the students and the results are invented for teaching purposes, and any resemblance to a real survey is coincidental. Read it as a model of structure and method, not as a report of real research.

## 15.1 The question and the questionnaire

The fictional study asks: 'How satisfied are undergraduate students with their university library, and does satisfaction differ by year of study?' The questionnaire has 14 items: one on frequency of use, one on purpose of use, five Likert-scale items on satisfaction with books, opening hours, staff, study space and computers, three closed questions on facilities, one open question asking for the single most wanted improvement, and two background items on year of study and subject. The questionnaire was piloted with 15 students, which led to two changes: the word 'provision' was replaced with 'books and computers', and a missing option, 'Not employed', was added to a background item.

## 15.2 Sampling and administration

The population was defined as all undergraduate students in the Faculty of Education, a total of 600 students. The target sample size was set using a standard sample-size table, and a stratified random sample was drawn, with equal numbers from each year of study. The questionnaire was sent online, with a short invitation explaining the purpose, the time needed (about five minutes), and that participation was voluntary and anonymous. Two reminders were sent. Of the 200 students invited, 120 completed the questionnaire, a response rate of 60 per cent. Final-year students were under-represented, and this is stated as a limit of the study.

## 15.3 Results

The results showed that 63 per cent of respondents were satisfied with the opening hours and 71 per cent with the books. Satisfaction with opening hours rose with year of study: 52 per cent of first-year students, 61 per cent of second-year students, and 78 per cent of final-year students. The open question produced a clear pattern: the most common request, from 38 respondents, was for longer evening opening hours, followed by more computers, requested by 24 respondents.

### 15.4 Conclusion and recommendations

The study concluded that students were generally satisfied with the library, but that satisfaction with opening hours was lower among first-year students, who used the evening hours less, and that the clearest demand was for longer evening opening. It recommended extending evening opening hours during term time, adding computers in the study area, and repeating the survey after one year to see whether satisfaction had changed. The report also noted the limits: a single faculty, a 60 per cent response rate, and under-representation of final-year students.

**Worked Example: What makes this example work**

Notice what the fictional survey does: it starts with one clear question, turns it into a list of variables, writes one item per variable, pilots the questionnaire and revises it, samples by a named method with a size taken from a standard table, reports the response rate with numbers, analyses with counts, percentages and cross-tabulation, and states its limits. Each of those steps comes from an earlier chapter of this guide. A real survey that follows the same steps, with the same discipline, will produce a report of the same shape.

[ ] The fictional survey follows the same steps this guide recommends.

[ ] The questionnaire is short, piloted and revised.

[ ] The sample size comes from a standard table, not from guesswork.

[ ] The response rate is reported with numbers.

[ ] The limits of the study are stated in the conclusion.

## Chapter 16: Conclusion

A good survey is not a difficult thing to design, but it is an easy thing to get wrong. The difference lies in discipline: start with one clear research question, turn it into a list of variables, write one good question per variable, test the questionnaire on real people before you use it, choose your sample honestly, report your response rate, clean your data carefully, and describe your results with numbers, tables and charts that a reader can understand in seconds.

If you follow the steps in this guide, your survey will stand up to examination — not because it is long or complicated, but because it is honest. It will show what you asked, whom you asked, how many answered, what they said, and what that means. And it will state plainly what it cannot show. That honesty is the mark of good research. Write the questionnaire carefully, pilot it properly, and let the respondents speak. Good luck — and get surveying.

# Suggested Reading

The following are general, well-known kinds of works that students commonly consult for survey research. Check your university library for the latest editions, and follow your department's guidance on referencing style.

- Standard survey research textbooks: widely used methodology texts that explain questionnaire design, sampling, response rates and the analysis of survey data, with

worked examples from education, business and the social sciences.

- General research methods textbooks: standard methodology texts used in most universities, covering research design, sampling, ethics and the reporting of research.
- Introductory statistics textbooks: standard works that explain descriptive statistics, cross-tabulation and the basic statistical tests used with survey data, suitable for students with no previous background in statistics.
- Style manuals: the official manuals of the major referencing styles (APA, Harvard, Chicago, MLA) are the authoritative source for citation and formatting rules — consult the edition your department requires.
- Your university's thesis regulations and template: always the first and final authority on structure, word count, formatting and submission requirements.
- Your supervisor's recommended reading list for your subject: the most relevant source of all, because it is chosen for your specific topic and method.
- Standard sample-size tables and calculators: the usual tools for choosing a sample size for a given population — ask your supervisor or your university library which one to use.